Randomized and low-rank approximation

Relative-error threshold for extremal Gaussian trace bounds

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because an explicit threshold must control extremal tails uniformly over spectra; community impact is sharper distribution-level trace-estimation confidence bounds.
Source: Hallman, Conjecture 3 together with Theorem 6.

Related auxiliary counterexamples — 2026-09-11

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The submitted Gamma-density examples refute the upper-mode assertion in Hallman Conjecture 1 and the upper-inflection assertion in Conjecture 2, with legally distinct augmentation indices. These are counterexamples to auxiliary assertions.

Remaining question: The complete relative Gaussian trace-tail probability chain in this entry is neither proved nor refuted. No revised sharp tail threshold is established; status remains Open.

Primary reference: complete authored PDF, standalone TeX, **Proposition 2.1 (with Proposition 3.1 for related evidence)**. Independent proof review · Authorship and submission record. Verification is independent agent review, not external human peer review or formal certification.

Problem statement

For a real symmetric $d \times d$ matrix D and integer $m \geq 1$, define the Gaussian trace estimator

$$T_m(D) = \frac{1}{m} \sum_{j=1}^{m} z_j^T D z_j, \qquad z_1, \dots, z_m \overset{\text{iid}}{\sim} N(0, I_d).$$

Let $A \neq 0$ be any real symmetric positive semidefinite $n \times n$ matrix, with $n \geq 1$, and set

$$\mu = \frac{\operatorname{tr}(A)}{\|A\|_2}, \qquad B_\mu = \frac{1}{\mu} \operatorname{diag}\left(I_{\lfloor \mu \rfloor}, \mu - \lfloor \mu \rfloor\right).$$

Here $\|\cdot\|_2$ is the spectral norm, $\mu \geq 1$, and a zero final diagonal entry may be retained. Let X have the Gamma distribution with shape and rate both $m\mu/2$. The shape/rate convention means that $\operatorname{Gamma}(\alpha, \beta)$ has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ for $x > 0$.

Conjecture. For every such A, every integer $m \geq 1$, and every $\varepsilon \geq 2/(m\mu)$, the complete comparison chain holds:

$$\begin{aligned}\Pr(|T_m(A) - \operatorname{tr}(A)| \geq \varepsilon \operatorname{tr}(A)) &\leq \Pr\left(|T_m(B_\mu) - 1| \geq \varepsilon\right) \\ &\leq \Pr(|X - 1| \geq \varepsilon).\end{aligned}$$

Each estimator uses Gaussian vectors of its own matrix dimension; only their distributions are compared.

Why it matters in numerical linear algebra

This would specify the tolerance range on which effective rank yields extremal, distribution-level confidence bounds for Gaussian trace estimation. It also applies to Frobenius-norm estimation through $\|C\|_F^2 = \operatorname{tr}(C^T C)$.

References

1. Eric Hallman, *Extremal bounds for Gaussian trace estimation*, arXiv:2411.15454v1 (2024), §5, Theorem 6 and Conjecture 3.
2. Alice Cortinovis and Daniel Kressner, *On Randomized Trace Estimates for Indefinite Matrices with an Application to Determinants*, FoCM 22 (2022), 875–903, Theorem 1.

Status check

Theorem 6 proves the comparisons beyond an unspecified threshold; Conjecture 3 supplies the explicit threshold above. On 2026-09-08 the arXiv record still listed only v1. Title, author, conjecture-number and 2025/2026 searches found no resolution. The author's later XTrace paper studies different estimators. This is a bounded check.

Audit — 2026-09-10

Rechecked Hallman, Theorem 6 and Conjecture 3; the record still lists only v1. Author, Gaussian-trace, and conjecture searches found no resolution. An unspecified larger threshold proves a weaker result and does not verify the displayed explicit threshold.